



Keycloak's Test Evolution: Your Guide to Mastering the New Test Framework

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KEYCLOAK DevDay 2026

Who's this guy?

- Simon Vacek, IBM employee
- Brno, Czech republic
- At Keycloak since July 2023
- On the new test framework almost since the beginning



Who's the crowd?

Raise your hand if you...

Have you been to KeyConf25?

The new Test Framework

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 - All in Java code
- Custom tailored for Keycloak
 - Easy access to realms, users, admin client, run code on the server, ...
 - Configure the Keycloak server, add providers, choose different database, ...

Who's the crowd?

Raise your hand if you...

Have you used the new framework?

Show me the code

Keycloak contributor perspective

The Basics

- `@KeycloakIntegrationTest`
- `@InjectRealm`, `@InjectClient`, ...
- Managed objects with a little extra
- Configuration in Java classes

Scenario 1: Failed login sends an email

- Set up an email event listener
- Perform login with wrong credentials
- Receive an email with an error message

Scenario 2: Change default signature algorithm

- For the duration of the test, change the algorithm in the master realm
- Get new access token
- Verify the token has the new algorithm in the header

Scenario 3: Disabled feature

- Disable the Impersonation feature on the server
- Try to impersonate a user
- Operation fails

Scenario 4: Verify required consent

- Set required consent for a test-app client
- Login with user credentials
- A page asking for consent to share data is displayed
- After clicking cancel, error is displayed

Who's the crowd?

Raise your hand if you...

Do you contribute to a Keycloak extension?

Show me the code

Custom project perspective

Custom provider

- Adding a REST endpoint /devday/hello
- Implementation of RealmResourceProvider
- Returns a text message

Configuration

Configuration

- SmallRye
- Sources:
 - System properties
 - Environment variables
 - `.env.test` file

Configuration

KC_TEST_SERVER

- embedded
 - Executes Keycloak within the JVM used to execute tests
- distribution (default)
 - Installs Keycloak from a zip file and starts as an external process
- remote
 - Connects to a running Keycloak process

Configuration

KC_TEST_SERVER=distribution

- KC_TEST_SERVER_REUSE=true
 - Server keeps running after tests finish
 - Faster test execution
- KC_TEST_SERVER_HOT_DEPLOY=true
 - Dependencies deployed from current available compiled classes

Configuration

KC_TEST_DATABASE

- H2 mem, file
- MariaDB
- MySQL
- MSSQL
- Oracle
- Postgres
- TiDB
- Remote

Configuration

- KC_TEST_BROWSER
- KC_TEST_LOG_FILTER
 - Filter out logs of successful tests
- And many more logging options

Session Summary

Summary

- Automatically managed instances
 - Realm, user, client, adminclient, oauthclient, webdriver, mailserver, pages, ...
- Resource configuration with builder pattern in Java code
- Configuration for DB, server, logging, browser

Read more

- Framework docs ([keycloak/test-framework/docs](https://keycloak.org/2020/11/11/test-framework/docs))
- GitHub Discussion ([Keycloak Test Framework - Full Support](https://github.com/keycloak/keycloak/discussions/11111))
- Session materials ([vaceksimon/DevDay26](https://vaceksimon.com/DevDay26))

Feedback on this session:



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